

Biopharma strategy needs regulatory focus



REJI K JOSEPH

Faculty member at Institute for Studies in Industrial Development, New Delhi

Policymakers must turn attention to comparative efficacy data and animal testing requirement waiver, easing enforcement of clinical trial regulation, and integrating AI into clinical trials

IN BUDGET 2026, Finance Minister Nirmala Sitharaman announced a plan to scale up manufacturing in seven strategic and frontier sectors, including biopharmaceuticals. The “Strategy for Healthcare Advancement through Knowledge, Technology and Innovation” in biopharma (Biopharma SHAKTI) has an allocation of ₹10,000 crore over the next five years to build the ecosystem for domestic production of biologics and biosimilars. The strategy has three components—establishing three National Institutes of Pharmaceutical Education and Research focused on biopharma; strengthening the Central Drugs Standard Control Organisation (CDSCO), the nodal agency overseeing regulations related to medicines, to meet global standards and approval timeframes; and creating a network of over 1,000 accredited clinical trial sites in India.

The strategy assumes great significance when biopharma (large molecule drugs) is gaining market share at an accelerated pace. In the global market, the share of biologics increased from 31% in 2018 to 42% in 2023 and is projected to outpace small molecule drugs (chemical-based drugs) by 2027. As biopharma products provide effective treatment for complex diseases like cancer, besides autoimmune diseases and rare genetic conditions, building up a sound production ecosystem in India is critical for its public health security and for taking advantage of emerging opportunities in the global market.

As regulations play an important role in the growth of this sector, the Budget ap-

tly includes the CDSCO as one of the components of the strategy. There are three areas of regulations where the immediate attention of policymakers is required.

One, waiving off the comparative efficacy data and animal testing requirements. Compared to generic medicines, biosimilars require substantially more data for market approval. While generics are approved based on pharma equivalence and bioequivalence data, biosimilars follow a totality-of-evidence approach requiring comparative analytical data involving pre-clinical and clinical studies to establish similarity with the reference biologic. This makes developing biosimilar costlier and more time-consuming, resulting in much higher prices. According to data published by the Generics and Biosimilars Initiative, while approval of generics costs \$2-5 million, takes two-three years and involves 20-50 patients, biosimilar approval takes \$100-200 million, eight-10 years, and 500 patients.

Now with the emergence of advanced analytical technologies, leading regulatory regimes such as the European Medicines Agency, US Food and Drug Administration (USFDA), and Health Canada are waiving off many of these data generation requirements, making biosimilar development more competitive. In 2023, Health Canada approved Samsung Bioepis’s biosimilar Pyzchiva without requiring com-

parative efficacy data. Countries are also doing away with animal study requirements as there are alternatives such as AI-based computational models to generate such data. Last year, the USFDA issued a draft guideline stating comparative efficacy studies may not be required for biosimilar nod. It has also announced plans to do away with animal study requirements.

The CDSCO is gradually moving in this direction. It issued a draft guideline in May 2025 to invite comments on a proposal to waive off much of the data requirement for approval of biosimilars. It is a progressive step, but the draft guideline has drawbacks. It does not seem to have fully aligned with the latest advances in technology and regulations as it has not fully waived off animal testing

requirements. It also suggests clinical trials would be waived off when sufficient evidence of biosimilarity exists, but what is sufficient evidence is left to the regulator’s discretion. The proposed regulation must be formalised early, incorporating the scientific principles behind regulatory innovations in leading regimes and minimising regulatory discretion for a more predictable and efficient approval process.

Two, streamlining the implementation of clinical trial regulation. Regulations concerning clinical trials, which account for about 40% of the pharma R&D budget and take three-six years, require an

overhaul in their implementation. The New Drugs and Clinical Trial Rules, 2019, were introduced with the objective of addressing malpractices in clinical trials in India. However, some critical lacunae in implementation have been pointed out, including lack of coordination among the three-tier committee system, ultimately resulting in India losing its appeal for clinical trials. India’s share in clinical trials has fallen from 11% in 2019 to 3% in 2025. Some of the leading innovative pharma firms have shifted their clinical trials out of India for reasons that include concerns over the implementation of regulations.

Three, integrating AI into clinical trials. AI has increasingly been used in drug discovery and development, including the clinical trials phase. Patient data is critical for developing AI models required for the pharma industry. Access to this data will also make clinical trials smoother, especially in identifying the right individuals. The Digital Personal Data Protection Act, 2023, mandates anonymisation of personal health data, a necessary step for privacy protection but one that adds short-term costs. Although India generates vast quantities of health-related data, much of it remains siloed and inaccessible, limiting its utility for AI-driven drug discovery and development.

The Budget has identified biopharma as one of the strategic sectors and laid down a strategy for its transformation. There are a few areas of regulations that require the immediate attention of policymakers to make the strategy work well.

A sound production ecosystem is critical for India's public health security and taking advantage of opportunities